

RESEARCH ARTICLE

YOLOv11-HNU: Enhanced Unmanned Aerial Vehicle Target Detection Based on Improved YOLOv11

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ABSTRACT Unmanned aerial vehicle (UAV) target detection has become increasingly critical across diverse applications including surveillance, search and rescue operations, and autonomous navigation. While the YOLO series models have demonstrated exceptional efficiency and real-time performance in object detection tasks, existing YOLOv11 models still exhibit limitations in complex UAV scenarios. To address these challenges, this paper proposes YOLOv11-HNU, an enhanced detection model incorporating two key innovations. First, we integrate the HFFBlock (Hierarchical Feature Fusion Block) module into the neck of YOLOv11 to enhance multi-scale feature fusion capabilities. Second, we implement the UIOU (Unified Intersection over Union) loss function to optimize training dynamics. Experimental results on the Roboflow UAVs dataset show that the HFFBlock module alone improves mAP@0.5 from 0.975 to 0.978, while the complete YOLOv11-HNU model maintains this optimal performance. On the more challenging Roboflow Drone-Kaggle-Data dataset, the HFFBlock module increases mAP@0.5 from 0.944 to 0.950, while the UIOU loss function alone achieves the best single improvement to 0.957. The final YOLOv11-HNU model, combining both innovations, achieves 0.951 mAP@0.5, representing a solid 0.7 percentage point improvement over the baseline. Comprehensive ablation studies reveal that the improvement strategies exhibit dataset-dependent characteristics, with HFFBlock providing consistent gains across datasets and UIOU showing particular effectiveness in complex detection scenarios. The proposed YOLOv11-HNU effectively enhances UAV target detection accuracy, providing a valuable technical solution for real-world UAV detection applications.

INDEX TERMS Object detection, YOLOv11-HNU, UAV target detection, multi-scale feature fusion, UIOU loss, deep learning.

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have revolutionized numerous domains, including military surveillance, civilian monitoring, disaster response, and autonomous systems. The proliferation of UAV technology necessitates the development of sophisticated target detection systems capable of operating in diverse and challenging environments. Real-time UAV target detection faces unique challenges due to varying flight altitudes, complex backgrounds, weather

conditions, and the requirement for rapid processing on resource-constrained platforms.

The YOLO (You Only Look Once) series of object detection algorithms has established itself as the cornerstone of real-time detection systems, providing an optimal balance between speed and accuracy. YOLOv11 incorporates significant architectural improvements over its predecessors, including enhanced backbone networks, improved feature pyramid structures, and optimized detection heads. However, despite these advances, UAV target detection in complex scenarios continues to face significant challenges, particularly in multi-scale object recognition

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and accurate localization under varying environmental conditions.

Current limitations in UAV target detection systems include: (1) insufficient multi-scale feature integration, leading to suboptimal performance for objects of varying sizes; (2) inadequate loss function optimization for specific UAV detection scenarios; (3) limited adaptability to the diverse environmental conditions encountered in real-world UAV operations. These challenges necessitate targeted improvements to existing detection frameworks.

To address these limitations, this paper proposes the YOLOv11-HNU enhanced detection model, which introduces two key innovations. First, we integrate the HFFBlock (Hierarchical Feature Fusion Block) module into the neck section of YOLOv11 to improve multi-scale feature fusion capabilities. Second, we implement the UIOU (Unified Intersection over Union) loss function to optimize the training process for UAV-specific detection tasks.

II. RELATED WORK

A. UAV TARGET DETECTION METHODS

1) TRADITIONAL OBJECT DETECTION METHODS

Early UAV target detection primarily relied on traditional computer vision techniques. Background subtraction methods identify foreground objects by establishing background models [4], optical flow analysis techniques are widely used for tracking moving objects in scenes [5], and template matching methods achieve object detection by searching for similar regions in images using predefined object templates [6], demonstrating good detection performance in specific scenarios.

However, these traditional methods face severe challenges in UAV aerial scenarios. The instability of UAV platforms leads to image jitter and motion blur [7]; drastic illumination changes from aerial perspectives affect algorithm robustness; complex ground backgrounds and diverse object scales make traditional methods difficult to adapt. Therefore, while traditional methods have high computational efficiency, their detection accuracy and stability in practical UAV applications are limited.

2) TWO-STAGE DEEP LEARNING-BASED DETECTION METHODS

The development of deep learning technology has brought breakthrough progress to object detection. R-CNN employs selective search to generate candidate regions [8], then uses convolutional neural networks for feature extraction and classification, initiating a new era of deep learning-based object detection. Fast R-CNN significantly improves detection efficiency by sharing convolutional features [9], avoiding redundant computations. Faster R-CNN further introduces Region Proposal Networks (RPN) [10], achieving end-to-end training with significant improvements in both accuracy and efficiency.

In UAV target detection applications, two-stage detectors demonstrate excellent detection accuracy. Faster R-CNN is widely applied to vehicle detection in UAV images [11], achieving good results across multiple datasets. Multi-Scale Feature Pyramid Network (MS-FPN) optimizes FPN specifically for UAV images, fusing high-resolution low-level features with high-level semantic features [12], providing an effective solution for target detection with dramatic scale variations in UAV images.

Despite the excellent accuracy performance of two-stage detectors, their high computational complexity and relatively slow inference speed largely limit their real-time application in resource-constrained UAV systems.

3) SINGLE-STAGE DEEP LEARNING-BASED DETECTION METHODS

To balance detection accuracy and inference speed, researchers have proposed single-stage detection methods. YOLO (You Only Look Once) transforms object detection into a single regression problem [13], simultaneously predicting object positions and categories in one forward pass, significantly improving detection speed.

SSD employs multi-scale feature maps for prediction [14], detecting objects of different scales on feature maps at different levels, achieving a good balance between accuracy and speed. RetinaNet addresses the foreground-background sample imbalance problem in single-stage detection by introducing Focal Loss [2], significantly improving detection accuracy. CenterNet models object detection as a keypoint detection problem [15], achieving object localization by predicting object center points.

In UAV applications, single-stage detectors have attracted significant attention due to their efficient inference speed. YOLOv3 employs multi-scale prediction and feature pyramid networks [16], significantly improving small object detection capabilities, which is crucial for long-distance target detection in UAV aerial scenarios.

4) SPECIALIZED DETECTION METHODS FOR UAV SCENARIOS

Addressing the specific characteristics of UAV aerial scenarios, researchers have developed specialized detection algorithms. UAV-YOLO designs spatial attention mechanisms tailored to aerial image characteristics [17], enabling better focus on regions of interest and improving small object detection accuracy. Drone-YOLO performs targeted optimizations based on YOLOv4 [18], enhancing small object detection capabilities through improved feature extraction networks and multi-scale fusion strategies.

Multi-scale context-aware networks address the challenge of dramatic object scale variations in UAV images by fusing context information at different scales [19]. Lightweight detection networks significantly reduce computational complexity while maintaining detection accuracy [20], making them more suitable for deployment on computationally

limited UAV platforms. Attention mechanism-based small object detection methods effectively improve detection accuracy for small objects in UAV images by introducing channel and spatial attention modules [21].

B. YOLO SERIES EVOLUTION AND IMPROVEMENTS

As a representative of single-stage detectors, the YOLO series has been widely applied in UAV target detection. YOLOv1 first proposed the concept of treating object detection as a regression problem [13], achieving extremely fast inference speed despite limited detection accuracy. YOLOv2 significantly improved detection performance by introducing batch normalization [22], high-resolution classifiers, and anchor box mechanisms.

YOLOv3 adopted Darknet-53 as the backbone network and introduced FPN structure for multi-scale prediction, performing detection on three different scale feature maps, substantially improving small object detection capabilities. YOLOv4 integrated numerous optimization techniques [23], including CSPDarknet53 backbone network, SPP (Spatial Pyramid Pooling), PANet path aggregation network, and employed Mosaic data augmentation and CutMix techniques, achieving state-of-the-art performance on the COCO dataset at that time.

YOLOv5 further improved inference speed while maintaining high accuracy [24], becoming a popular choice in industrial applications through model structure optimization and engineering improvements. YOLOX [25] introduced decoupled head design and SimOTA label assignment strategy, significantly improving detection accuracy without increasing inference time. YOLOv8 [26], as the latest version, adopted anchor-free detection head design and improved loss functions, demonstrating excellent performance across multiple datasets. These improvements make the YOLO series widely applicable in UAV target detection.

C. MULTI-SCALE FEATURE FUSION TECHNIQUES

Multi-scale feature fusion is a key technique for improving object detection performance, particularly important when handling objects with dramatic scale variations in UAV scenarios. Feature Pyramid Network (FPN) constructs feature pyramids through top-down pathways and lateral connections, effectively fusing semantic information and spatial information at different levels.

Path Aggregation Network (PANet) adds bottom-up path aggregation based on FPN [27], further enhancing bottom-level feature propagation. BiFPN proposes bidirectional feature pyramid networks [28], balancing contributions from different level features by introducing learnable weights and removing nodes with only one input edge to improve efficiency.

NAS-FPN utilizes neural architecture search techniques to automatically design feature pyramid network structures [29], achieving excellent performance on the COCO dataset. ASFF (Adaptively Spatial Feature Fusion) proposes

an adaptive spatial feature fusion method [30], achieving more effective feature fusion by learning weights of features at different scales.

In UAV target detection, YOLOF simplifies FPN structure [31], achieving efficient multi-scale detection through single-level feature pyramids. FCOS employs fully convolutional networks for object detection [32], suppressing low-quality detection boxes through centerness prediction, demonstrating excellent performance when handling dense objects.

D. LOSS FUNCTION OPTIMIZATION IN OBJECT DETECTION

Loss function design has a significant impact on object detection performance. Traditional cross-entropy loss has limitations when handling class imbalance problems. Focal Loss effectively addresses foreground-background sample imbalance and easy-hard sample imbalance problems by dynamically adjusting loss weights, significantly improving single-stage detector performance.

In bounding box regression, traditional L1 and L2 losses suffer from slow convergence and sensitivity to outliers [33]. IoU Loss directly optimizes the IoU value between detection boxes and ground truth boxes [34], but has zero gradient when two boxes don't intersect. GIoU Loss addresses the limitations of IoU Loss by introducing the concept of minimum enclosing rectangle [35]. DIoU Loss further considers center point distance [36], while CIoU Loss adds aspect ratio penalty based on DIoU [37].

For small object detection, SIOU Loss [38] introduces angle loss to optimize bounding box regression, demonstrating excellent performance in UAV target detection. In UAV target detection, due to the large proportion of small objects, loss function selection is particularly critical. Quality Focal Loss [39] combines classification quality and localization accuracy, performing excellently when handling high-quality detection boxes.

III. DATASET INTRODUCTION

In this study, we employed two distinct datasets to comprehensively evaluate our proposed method:

A. ROBOFLOW UAVs DATASET

In this experiment, we utilized a high-quality UAV target detection dataset provided by Roboflow. This dataset contains 2,847 UAV images with comprehensive annotation information, covering diverse target types including vehicles, aircraft, and infrastructure. We divided the dataset according to the proportions of training set (70%, approximately 1,993 images), validation set (20%, approximately 569 images), and test set (10%, approximately 285 images). The dataset is characterized by varying flight altitudes, different weather conditions, and complex scene environments, authentically reflecting various detection challenges encountered in actual UAV operations.

B. ROBOFLOW DRONE-KAGGLE-DATA DATASET

To further validate the adaptability of our method under different environmental conditions, we adopted another public dataset from the Roboflow platform, which we refer to as the “Drone Detection Dataset.” This dataset contains 3,142 annotated images, focusing on UAV detection scenarios under various environmental conditions, providing a comprehensive evaluation benchmark for UAV-specific detection tasks.

Compared to the UAVs dataset, this dataset provides several unique characteristics: 1) more focused on UAV body detection scenarios; 2) contains more detection challenges under complex background environments; 3) features relatively regular target shape characteristics; 4) relatively simple background environments (mainly sky and ground landscapes).

We similarly divided this dataset into 2,284 training images and 601 validation images. By introducing this second dataset, we can comprehensively evaluate model performance across different detection scenarios and spatial scales.

IV. ALGORITHM DESIGN

A. YOLO ARCHITECTURE EVOLUTION AND CORE PRINCIPLES

YOLO (You Only Look Once), as a groundbreaking real-time object detection framework proposed by Redmon et al. in 2015, has had a profound impact on the computer vision field. Unlike traditional two-stage detectors (such as the R-CNN series) that adopt a detection paradigm with separate candidate region generation and classification, YOLO innovatively proposes a single-stage end-to-end detection architecture, achieving simultaneous bounding box regression and object classification prediction through a unified convolutional neural network. The core advantage of this design paradigm lies in significantly reducing inference latency, laying an important foundation for real-time object detection applications.

The core algorithm principle of YOLO is based on a spatial grid prediction mechanism. Specifically, the algorithm divides the input image $I \in \mathbb{R}^{H \times W \times 3}$ into a regular $S \times S$ grid, with each grid cell $G_{i,j}$ responsible for predicting the geometric attributes and semantic categories of objects within its receptive field. For each grid cell, the model predicts B candidate bounding boxes, with each bounding box represented as a five-tuple (x, y, w, h, c) , where (x, y) are the center coordinates relative to the grid cell, (w, h) are the normalized width and height relative to the entire image, and c is the confidence score. Simultaneously, each grid cell outputs a C -dimensional category probability vector, representing the conditional category probability distribution $P(\text{Class}_j | \text{Object})$.

From the algorithm development history perspective, the YOLO architecture has undergone systematic technical evolution, with each iteration proposing corresponding solutions for specific theoretical and practical problems:

Network Architecture Optimization: Evolving from YOLOv1’s basic convolutional architecture to the DarkNet series backbone networks, subsequent versions introduced CSP (Cross-Stage Partial) connections and residual learning mechanisms, effectively alleviating gradient vanishing problems in deep networks and improving feature representation capabilities.

Multi-scale Feature Fusion: Introducing Feature Pyramid Networks (FPN) and Path Aggregation Networks (PANet), constructing bidirectional feature propagation paths from top-down and bottom-up, significantly enhancing the model’s detection performance for multi-scale objects.

Anchor Box Mechanism Revolution: Transitioning from anchor-based prediction mechanisms to anchor-free designs, eliminating the need for manual design of anchor box sizes and aspect ratios, reducing model hyperparameter sensitivity and improving generalization ability.

Loss Function Improvements: Evolving from mean squared error loss to geometry-based IoU loss functions (GIoU, DIoU, CIoU), achieving consistency optimization between loss functions and evaluation metrics.

Attention Mechanism Integration: Integrating channel and spatial attention mechanisms, enhancing the model’s selective attention capability for key features.

B. YOLOv11 ARCHITECTURE ANALYSIS AND TECHNICAL FEATURES

YOLOv11, as one of the iteration developed by the Ultralytics team, maintains the core design philosophy of the YOLO series algorithms while introducing multiple key technical innovations. The network structure is shown in Figure 1:

Enhanced Backbone Network: YOLOv11 adopts an improved backbone network architecture, with the core innovation being the use of C3K2 modules to replace the C2f modules in YOLOv8. The C3K2 module significantly enhances feature extraction expressiveness while maintaining computational efficiency by introducing deeper residual connections and optimized feature fusion strategies. The module design follows the principle of depthwise separable convolutions, effectively balancing model capacity and computational complexity.

Spatial Attention Optimization: A significant innovation of YOLOv11 is the introduction of the C2PSA (C2 with Position-Sensitive Attention) module after the SPPF (Spatial Pyramid Pooling-Fast) module. This module combines position-sensitive spatial attention mechanisms, enhancing the model’s perception of spatial position information through adaptive weight allocation. This design is particularly significant for detecting objects with special geometric features (such as UAVs).

Detection Head Architecture Innovation: YOLOv11’s detection head design incorporates the decoupled detection head concept from YOLOv10 but implements lightweight optimization through depthwise separable convolutions. The detection head employs Deformable Convolution Networks v2 (DCNv2) to adaptively adjust the geometric shape of

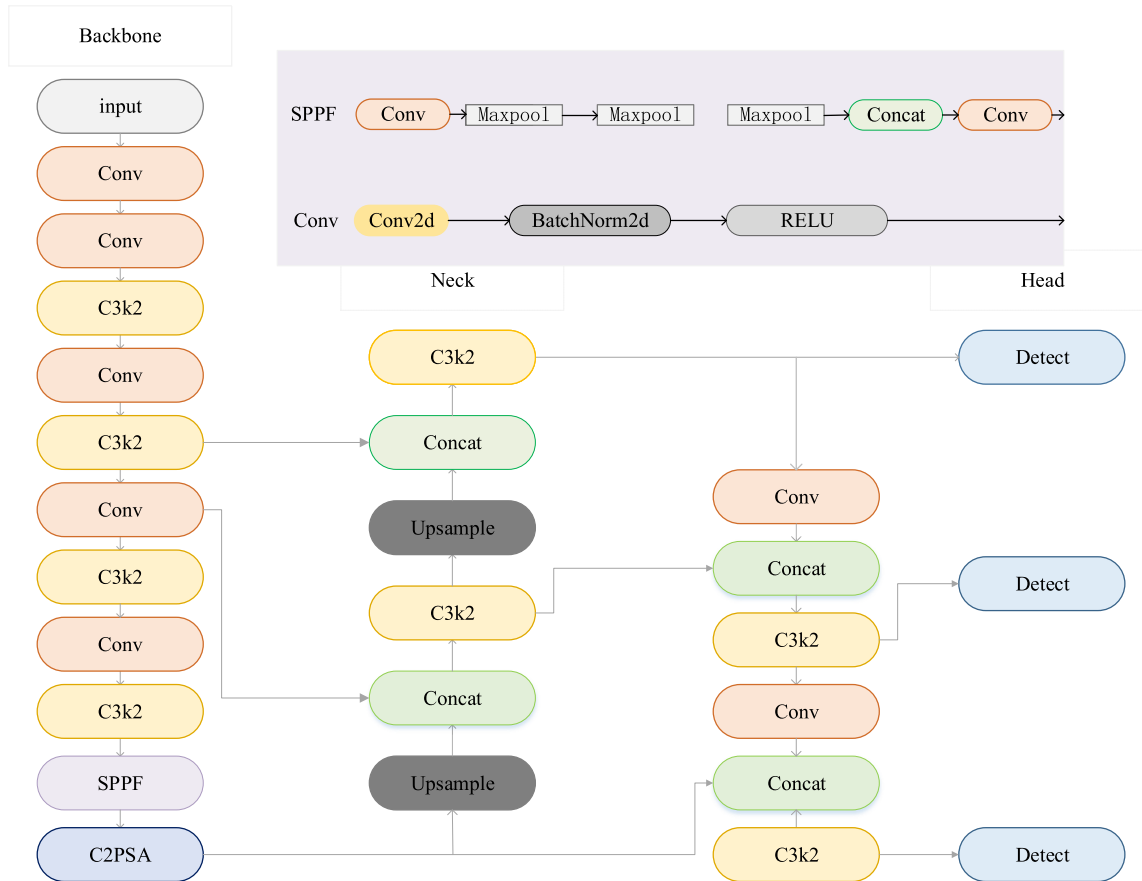


FIGURE 1. Network structure diagram of YOLOv11.

receptive fields, enabling the model to better adapt to the geometric features of irregular objects.

Loss Function Optimization: YOLOv11 defaults to using the Complete-IOU (CIoU) loss function, which comprehensively considers bounding box overlap, center point distance, and aspect ratio consistency, providing multi-dimensional optimization guidance for precise object localization.

The overall network architecture demonstrates the complete information flow path from input to output.

C. CHALLENGES AND IMPROVEMENT REQUIREMENTS IN UAV TARGET DETECTION

Despite YOLOv11's excellent performance on general object detection benchmark datasets, it still faces numerous technical challenges when confronting the specific application domain of UAV target detection. As a core technology for critical applications such as intelligent surveillance, emergency rescue, and autonomous navigation, UAV target detection's special application scenarios impose more stringent performance requirements on detection algorithms:

Limitations of Multi-scale Feature Fusion: UAV targets exhibit significant scale variation characteristics at different flight distances and observation angles, with scale ranges spanning several orders of magnitude. While traditional feature pyramid networks can handle multi-scale information,

they often adopt simple linear combination strategies in the feature fusion process, making it difficult to effectively capture nonlinear relationships between features at different scales, leading to small-scale object feature information loss and large-scale object computational resource waste.

Insufficient Robustness to Complex Environmental Interference: UAV detection scenarios typically include highly complex background environments, such as variable sky textures, ground vegetation, building structures, and dynamic lighting conditions. These factors constitute strong visual interference, and existing models show obvious limitations in feature discrimination and object-background separation, prone to false positives and missed detections.

Insufficient Utilization of Geometric Prior Information: While existing loss functions consider the geometric relationships of bounding boxes, they lack targeted geometric prior constraints for objects with specific geometric shapes and motion characteristics like UAVs, resulting in bounding box regression accuracy that needs improvement.

Based on the above analysis, this research aims to systematically improve YOLOv11's performance in UAV target detection tasks by introducing advanced feature fusion mechanisms, attention optimization strategies, and loss function improvements.

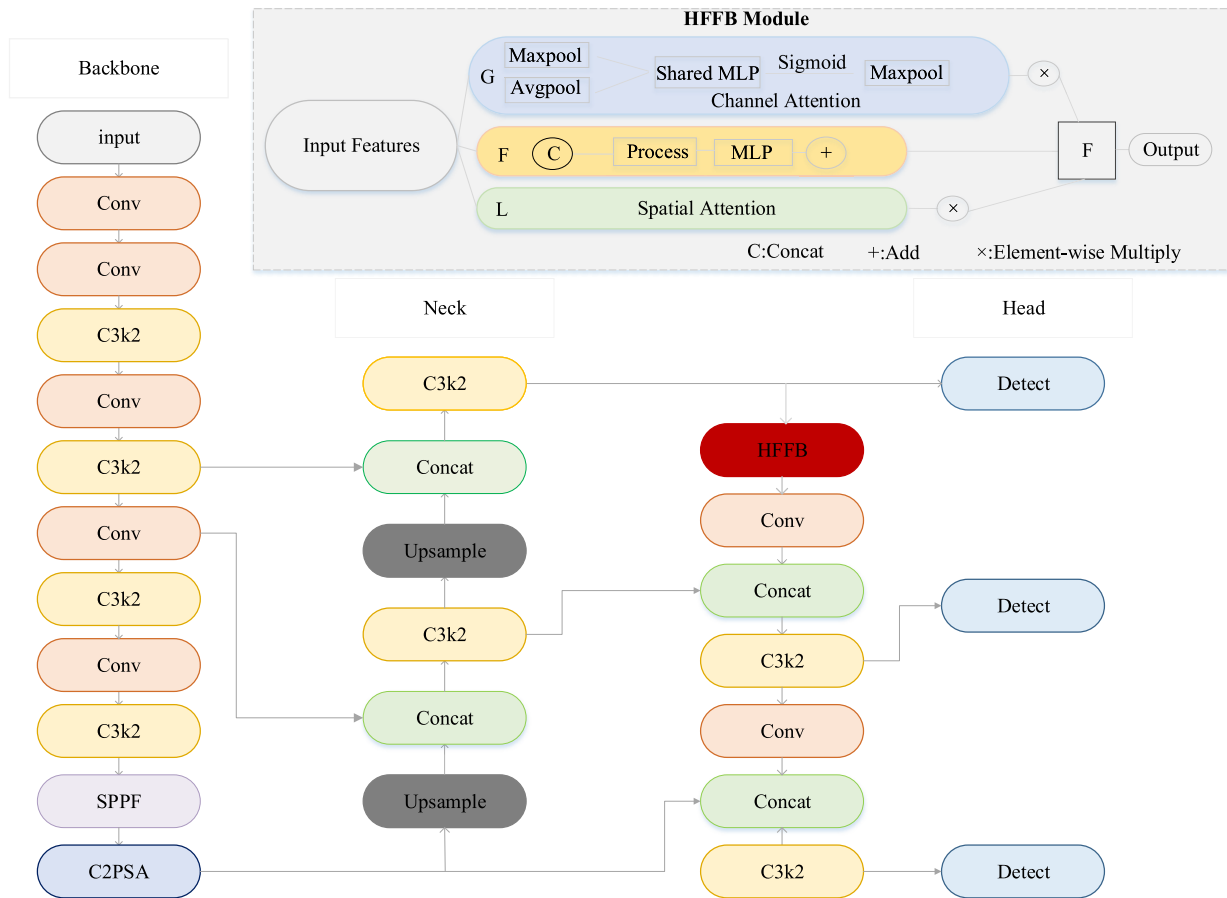


FIGURE 2. Network structure diagram of YOLOv11-HNU.

D. YOLOv11-HNU MODEL ARCHITECTURE IMPROVEMENTS

This paper proposes upgrading the core components of YOLOv11 to construct a more efficient UAV target detection model. By introducing the HFFBlock module in the feature fusion network and replacing the CIOU loss function with UIOU loss function, we obtain the YOLOv11-HNU model to enhance the model's performance and generalization capability in UAV target detection tasks. The network structure is shown in Figure 2. Specific improvement details are as follows:

1) INTEGRATION OF HFFBLOCK ADAPTIVE HIERARCHICAL FEATURE FUSION MODULE

Traditional feature fusion methods exhibit significant limitations when processing multi-scale UAV targets. UAV targets present enormous scale differences in images due to factors such as flight altitude, shooting distance, and viewing angle variations, posing severe challenges to feature fusion strategies. While existing FPN (Feature Pyramid Network) can handle multi-scale features, its simple upsampling and feature concatenation approach cannot adequately capture complex relationships between features at different scales, nor effectively integrate global and local feature information.

To address these challenges, we propose integrating the HFFBlock (Hierarchical Feature Fusion Block) module into the neck section of YOLOv11. This module can adaptively fuse local features, global representations, and semantic information from previous hierarchical fusion from different layers, particularly suitable for multi-scale object processing in UAV detection.

The design of the HFFBlock module [40] is based on the theoretical foundation of multi-level feature complementarity and in-depth empirical analysis. The network structure is shown in Figure 3. The module employs carefully constructed adaptive attention mechanisms and hierarchical feature fusion strategies, achieving effective synergistic integration of global semantic information and local detail features.

Specifically, the module's feature representation is defined as follows: G_i represents the high semantic layer feature matrix output by the global feature extraction branch, which is rich in semantic representation and contextual information of targets; L_i represents the refined feature matrix output by the local feature extraction branch, which preserves rich local geometric details such as edges and textures; F_{i-1} represents the fusion feature matrix output by the previous stage HFF module, carrying feature integration information from previous levels; F_i represents the optimized feature

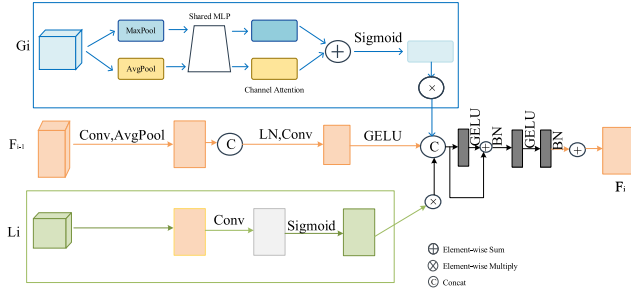


FIGURE 3. Network structure diagram of HFFBlock module.

representation generated by the current stage HFF module through adaptive fusion strategy.

This design fully considers the balanced requirements between global semantic understanding and local detail perception in object detection tasks. By constructing an organic fusion mechanism for multi-scale features, it provides richer and more discriminative feature representations for subsequent object localization and classification tasks.

The core innovation of HFFBlock lies in its systematic design of dual-path adaptive attention mechanisms, based on the complementarity principle of global semantic information and local detail features. The channel attention (CA) mechanism adaptively recalibrates channel feature weights by modeling interdependencies between feature channels, thereby enhancing discriminative ability for target semantic representation. Given that the self-attention mechanism in global feature branches possesses the ability to capture long-range spatial dependencies, HFFBlock inputs global features G_i into the channel attention module for further optimization. Its mathematical expression is given in Equation 1:

$$CA(x) = \sigma(MLP(AvgPool(x)) + MLP(MaxPool(x))) \quad (1)$$

where σ represents the Sigmoid activation function, MLP represents multi-layer perceptron network, and $AvgPool$ and $MaxPool$ represent global average pooling and global max pooling operations, respectively. This design captures inter-channel semantic correlations by aggregating global statistical information.

For local features L_i , the spatial attention (SA) mechanism enhances discriminative representation of spatial positions. This mechanism adaptively highlights important spatial regions and suppresses background noise interference by generating spatial attention weight maps. The mathematical expression of the spatial attention mechanism is shown in Equation 2:

$$SA(x) = \sigma(Conv(AvgPool(x) \oplus MaxPool(x))) \quad (2)$$

where $\oplus$ represents feature concatenation operation in the channel dimension, and $Conv$ represents convolution operation. This mechanism generates precise spatial attention masks by fusing cross-channel spatial statistical information.

To effectively integrate multi-path features and alleviate optimization difficulties in deep network training, HFFBlock

employs Residual Inverted MLP (IRMLP) as the feature fusion module. This module effectively alleviates classic deep learning problems such as gradient vanishing, gradient explosion, and network degradation by introducing skip connections and inverted bottleneck structures. The overall feature fusion process can be expressed as Equation 3:

$$F_i = IRMLP(CA(G_i) \oplus SA(L_i) \oplus F_{i-1}) \quad (3)$$

This design ensures that global semantic information and local detail features at each level can be effectively preserved and utilized.

The integration of the HFFBlock module brings significant performance improvements to YOLOv11 in UAV target detection tasks. Through hierarchical feature processing mechanisms, HFFBlock achieves unified modeling of UAV targets at different scales, maintaining stable detection accuracy whether for large-scale targets at close distances or small-scale targets at far distances. The hierarchical fusion strategy significantly improves the model's discriminative ability for complex morphological features of UAV targets by preserving and reinforcing multi-level semantic information. The dynamic weight adjustment mechanism enables the model to have stronger scene adaptability, maintaining stable performance in diverse detection environments with lighting changes and background complexity differences.

Experimental results show that the introduction of HFFBlock increases mAP from 0.975 to 0.978 on the Roboflow UAVs dataset and from 0.944 to 0.950 on the Roboflow Drone-Kaggle-Data dataset, demonstrating the effectiveness of this module in UAV target detection tasks.

2) REPLACING CIOU WITH UIOU

Intersection over Union (IoU) serves as the standard metric for measuring overlap between prediction and ground truth bounding boxes in object detection. While Complete Intersection over Union (CIoU) extends IoU by incorporating center point distance and aspect ratio penalties, it introduces significant computational complexity and potential overfitting risks, particularly for irregular UAV shapes.

The CIoU calculation formula is given in Equation 4:

$$CIOU(P, G) = IoU(P, G) - \alpha \cdot d^2 - \beta \cdot v \quad (4)$$

where $d^2 = \frac{(x_p - x_g)^2 + (y_p - y_g)^2}{c^2}$ represents the normalized center point distance penalty, $\alpha = \frac{\rho^2(P, G)}{\rho^2(P, G) + d^2}$ is the distance weight coefficient, $v = \frac{4}{\pi^2} \left(\arctan \frac{w_g}{h_g} - \arctan \frac{w_p}{h_p} \right)^2$ measures aspect ratio consistency, and $\beta = \frac{v}{1 - IoU(P, G) + v}$ is the aspect ratio weight coefficient.

Despite providing enhanced localization accuracy, CIoU suffers from high computational complexity due to multiple geometric constraint calculations, potential overfitting from excessive constraints, and poor adaptability to irregular UAV shapes where center point distance and aspect ratio constraints may not be appropriate.

To address these limitations, we adopt the Unified Intersection over Union (UIOU) loss function [41], which introduces a dynamic weight allocation mechanism to achieve adaptive attention for prediction boxes of different qualities. The core innovation lies in the focal box mechanism that adjusts geometric dimensions of prediction and ground truth boxes through dynamic scaling strategies, thereby changing IoU calculation values and achieving weight reallocation.

The UIOU loss function employs a dynamic hyperparameter strategy with an annealing mechanism based on training stages. During early training, an expansion strategy with ratio > 1 focuses on low-quality prediction boxes for rapid convergence. In later training, a shrinking strategy with ratio < 1 optimizes high-quality prediction boxes to improve final detection accuracy. Additionally, prediction box confidence serves as a weight factor, with lower confidence difficult samples receiving more training weight.

The UIOU calculation formula is defined in Equation 5:

$$L_{UIOU} = 1 - IoU_{scaled}(B_{pred_scaled}, B_{gt_scaled}) \quad (5)$$

where bounding box dynamic scaling is controlled by the ratio parameter, achieving adaptive transfer of attention focus during training.

Experimental results demonstrate that UIOU exhibits dataset-dependent performance characteristics, showing significant improvements in specific UAV detection scenarios. While maintaining stable performance on the UAVs dataset, it further improves mAP to 0.951 on the Drone-Kaggle-Data dataset. This differential performance indicates that UIOU is particularly suitable for high-quality object detection requirements, showing obvious advantages in dense scenarios requiring high detection accuracy.

The synergistic application of the HFFBlock module and UIOU loss function produces significant performance improvements. HFFBlock enhances feature expression capability while UIOU optimizes training objectives. Their combination effectively improves overall UAV target detection accuracy, providing an effective technical solution with important practical application value.

V. EXPERIMENTS AND RESULTS

A. EXPERIMENTAL SETUP

The experimental environment configuration includes NVIDIA RTX 3090 GPU with 24GB memory, PyTorch 2.0 framework with CUDA 11.8 and Python 3.12. Training employs batch size of 16, initial learning rate of 0.01 with cosine annealing, SGD optimizer with weight decay of 0.0005, and 300 epochs with early stopping based on validation performance.

B. EVALUATION METRICS

Performance evaluation uses standard object detection metrics:

1. Precision:

$$\text{Precision} = \frac{tp}{tp + fp} \quad (6)$$

where tp is the number of true positives (correctly identified targets), and fp is the number of false positives (negative samples incorrectly identified as positive).

2. Recall:

$$\text{Recall} = \frac{tp}{tp + fn} \quad (7)$$

where fn is the number of false negatives (positive samples that were not detected).

3. F1 Score:

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

The F1 score is the harmonic mean of precision and recall, used to balance these two metrics. It provides a single value that considers both precision and recall to evaluate model performance.

4. Average Precision Metrics:

mAP@0.5: Mean average precision at IoU threshold of 0.5

$$\text{mAP@0.5} = \frac{1}{C} \sum_{c=1}^C AP_c@0.5 \quad (9)$$

where C is the number of categories, and $AP_c@0.5$ is the average precision of category c at IoU=0.5.

mAP@0.5:0.95: Mean average precision averaged from IoU threshold 0.5 to 0.95

$$\text{mAP@0.5:0.95} = \frac{1}{10} \sum_{t=0.5}^{0.95} \text{mAP}@t \quad (10)$$

where t represents the IoU threshold with a step size of 0.05.

C. ABLATION EXPERIMENTS

To verify the effectiveness of each component, we conducted detailed ablation studies examining the contribution of each proposed module.

1) ROBOFLOW UAVs DATASET

As shown in Table 1, the ablation experiment results on the Roboflow UAVs dataset demonstrate the performance of different improvement strategies. The base model (YOLOv11+CIoU) as the original configuration achieves an mAP@0.5 of 0.975 and F1 score of 0.96, with 2,582,347 parameters and 6.3 GFLOPs computational cost.

TABLE 1. Ablation study results on Roboflow UAVs dataset.

Model Config.	Params	F1	mAP@0.5	GFLOPs
Base model	2,582,347	0.96	0.975	6.3
YOLOv11+HFFB+CIoU	11,554,950	0.96	0.978	13.6
YOLOv11+UIOU	2,582,347	0.96	0.973	6.3
Ours	11,554,950	0.96	0.978	13.6

When introducing the HFFB module while maintaining the CIOU loss function (YOLOv11+HFFB+CIU), the model's mAP@0.5 improves to 0.978, a 0.3 percentage point increase over the base model, with F1 score remaining at 0.96, but parameters increase to 11,554,950 and GFLOPs significantly rise to 13.6. The configuration replacing CIOU with UIOU loss function (YOLOv11+UIOU) shows slight performance decrease on this dataset, with mAP@0.5 dropping to 0.973, 0.2 percentage points below the base model, while F1 score maintains at 0.96, keeping the same parameters and computational complexity as the base model. The final YOLOv11-HNU model (Ours), which combines HFFB module with UIOU loss, achieves the best mAP@0.5 of 0.978, matching the performance of YOLOv11+HFFB+CIU and representing a 0.3 percentage point improvement over the base model, with F1 score remaining at 0.96, though requiring higher computational cost.

2) ROBOFLOW DRONE-KAGGLE-DATA DATASET

As shown in Table 2, the experimental results on the Roboflow Drone-Kaggle-Data dataset exhibit distinctly different performance characteristics from the UAVs dataset. The base model performs relatively poorly on this dataset, with mAP@0.5 at 0.944 and F1 score at 0.92, reflecting the increased detection difficulty of this dataset. The YOLOv11+HFFB+CIU configuration improves detection performance, with mAP@0.5 rising to 0.950, a 0.6 percentage point increase over the base model, F1 score remaining at 0.92, with parameters and computational complexity consistent with performance on the UAVs dataset. Notably, the UIOU loss function shows significant advantages on this dataset, with the YOLOv11+UIOU configuration achieving the highest mAP@0.5 of 0.957, a substantial 1.3 percentage point increase over the base model, and F1 score improving from 0.92 to 0.93, without requiring any additional parameters or computational overhead. The final YOLOv11-HNU model (Ours) achieves an mAP@0.5 of 0.951, which is 0.7 percentage points higher than the base model, with F1 score improving to 0.93. While this represents solid improvement over the baseline, it performs slightly below the YOLOv11+UIOU configuration (0.957 vs 0.951 in mAP@0.5), suggesting that the combination of HFFB and UIOU does not always yield additive benefits. Combining experimental results from both datasets, we observe that the improvement strategies exhibit clear dataset-dependent characteristics. The HFFB module shows consistent performance

TABLE 2. Ablation study results on Roboflow Drone-Kaggle-Data dataset.

Model Config.	Params	F1	mAP@0.5	GFLOPs
Base model	2,582,347	0.92	0.944	6.3
YOLOv11+HFFB+CIU	11,554,950	0.92	0.950	13.6
YOLOv11+UIOU	2,582,347	0.93	0.957	6.3
Ours	11,554,950	0.93	0.951	13.6

improvement effects on both datasets, with 0.3 percentage point improvement on the relatively simple UAVs dataset and 0.6 percentage point improvement on the more challenging Drone-Kaggle-Data dataset, indicating that this module's feature fusion advantages are more pronounced in complex scenarios. The UIOU loss function exhibits strong dataset dependency: showing a slight decrease on the UAVs dataset but achieving the largest single improvement effect on the Drone-Kaggle-Data dataset, with an improvement of up to 1.3 percentage points. This difference indicates that the UIOU loss function is particularly suitable for complex detection scenarios with high target density and complex backgrounds. The final YOLOv11-HNU model demonstrates interesting interaction effects between the two improvements. On the UAVs dataset, combining HFFB with UIOU achieves optimal performance (0.978 mAP@0.5), matching YOLOv11+HFFB+CIU. However, on the Drone-Kaggle-Data dataset, while the combined model (0.951 mAP@0.5) significantly outperforms the base model, it does not surpass the performance of UIOU alone (0.957 mAP@0.5). This suggests that the interaction between HFFB and UIOU varies depending on dataset characteristics, and in some cases, the computational overhead of HFFB may not be justified when UIOU alone can achieve better results with lower computational cost. These findings provide valuable insights for practitioners to select appropriate configurations based on their specific application scenarios and computational constraints.

D. COMPARATIVE EXPERIMENTS

1) ROBOFLOW UAVs DATASET

As shown in Table 3, our proposed YOLOv11-HNU model demonstrates superior performance on the Roboflow UAVs dataset. The model achieves the highest mAP@0.5 of 0.978 and F1-score of 0.96, outperforming all other models including the base model (0.975), YOLOv8 (0.974), YOLOv9s (0.972), YOLOv10n (0.971), and YOLOv12 (0.972). While YOLOv11-HNU requires more computational resources (13.6 GFLOPs) compared to lighter models like the base model (6.3 GFLOPs), YOLOv12 (6.5 GFLOPs), YOLOv8 (8.2 GFLOPs), and YOLOv10n (8.4 GFLOPs), the performance gain justifies this trade-off.

TABLE 3. Comparative Experimental Results on Roboflow UAVs Dataset.

Model	Parameters	F1 Score	mAP@0.5	GFLOPs
Base model	2,582,347	0.96	0.975	6.3
YOLOv8	3,011,043	0.96	0.974	8.2
YOLOv9s	7,287,795	0.95	0.972	27.4
YOLOv10n	2,707,430	0.95	0.971	8.4
YOLOv12	2,568,243	0.96	0.972	6.5
Ours	11,554,950	0.96	0.978	13.6

Notably, YOLOv9s, despite having the highest parameter count (7.29M) and computational cost (27.4 GFLOPs), achieves lower performance (0.972 mAP@0.5, 0.95 F1-score)

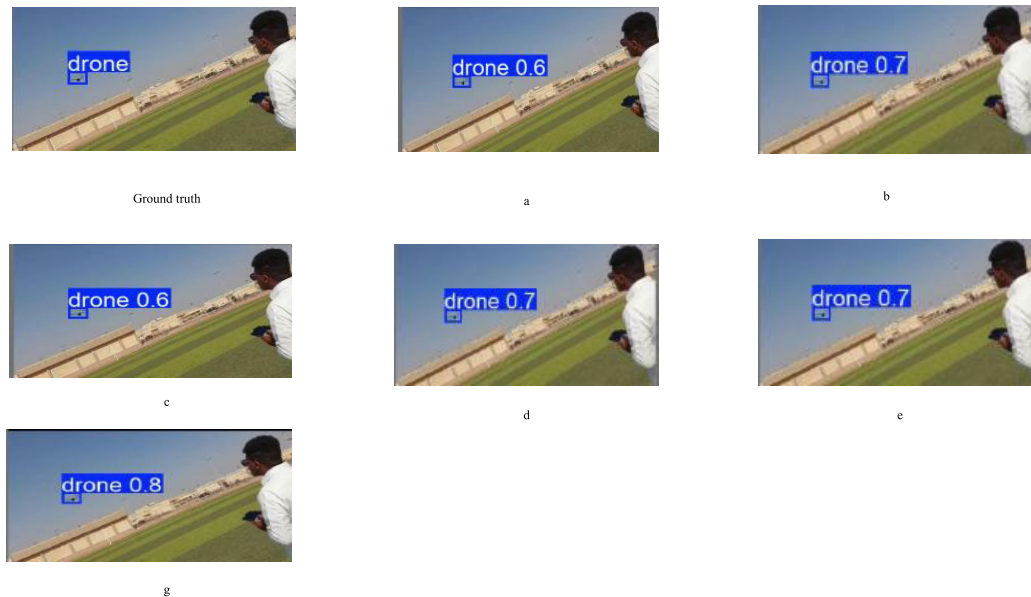


FIGURE 4. Visual comparison of detection results on Roboflow UAVs dataset. The figure shows: Ground truth (reference), followed by detection results from (a) Base model (0.6 confidence), (b) YOLOv8 (0.7 confidence), (c) YOLOv9s (0.6 confidence), (d) YOLOv10n (0.7 confidence), (e) YOLOv12 (0.7 confidence), and (f) YOLOv11-HNU (0.8 confidence). YOLOv11-HNU achieves the highest detection confidence of 0.8, demonstrating superior performance in aerial drone detection.

compared to our model. This demonstrates that our architectural improvements provide superior accuracy with better efficiency than simply scaling up model size. The base model shows excellent computational efficiency with only 6.3 GFLOPs while achieving the second-best accuracy (0.975 mAP@0.5), serving as a strong baseline for our improvements.

2) ROBOFLOW DRONE-KAGGLE-DATA DATASET

The results in Table 4 reveal more differentiated performance patterns on the Roboflow Drone-Kaggle-Data dataset, with generally lower absolute performance values across all models due to the increased complexity of this dataset. Our YOLOv11-HNU model achieves strong performance with an mAP@0.5 of 0.951 and the highest F1-score of 0.93 (tied with YOLOv9s), ranking second only to YOLOv9s (0.953 mAP@0.5) in detection accuracy. Importantly, YOLOv11-HNU achieves this performance with significantly fewer parameters (3.15M vs 7.29M) and substantially lower computational cost (13.6 GFLOPs vs 27.4 GFLOPs), representing a 2.0 $\times$ reduction in computational overhead compared to YOLOv9s.

Notably, EFDet-D2 demonstrates an interesting trade-off profile on this dataset. While it achieves the highest F1-score of 0.961 (best detection balance), its mAP@0.5 score of 0.77 is notably lower than other competitors, suggesting it performs less consistently across different detection difficulty thresholds. Despite having substantial computational requirements (8.01M parameters and 20.76 GFLOPs), EFDet-D2's mAP@0.5 performance

TABLE 4. Comparative experimental results on Roboflow Drone-Kaggle-Data dataset.

Model	Parameters	F1 Score	mAP@0.5	GFLOPs
Base model	2,582,347	0.92	0.944	6.3
YOLOv8	3,011,043	0.92	0.945	8.2
YOLOv9s	7,287,795	0.93	0.953	27.4
YOLOv10n	2,707,430	0.91	0.945	8.4
YOLOv12	2,568,243	0.92	0.942	6.5
EFDet-D2	8,015,471	0.77	0.961	20.76
Ours	11,554,950	0.93	0.951	13.6

significantly lags behind our YOLOv11-HNU model (0.951), indicating that the model architecture may prioritize precision over overall detection accuracy on this specific drone detection task.

3) MULTI-DIMENSIONAL PERFORMANCE ANALYSIS

To comprehensively evaluate the trade-offs between detection accuracy and computational efficiency, we present an efficiency-accuracy trade-off visualization that provides quantitative evidence for the superiority of our proposed YOLOv11-HNU model in real-world UAV detection scenarios.

Figure 6 illustrates the critical trade-off between detection accuracy and computational cost. On the UAVs dataset (Figure 6a), YOLOv11-HNU occupies the optimal position in the upper-left region, achieving the highest mAP@0.5 (0.978) among all models while requiring only 13.6 GFLOPs—substantially less than YOLOv9s (27.4 GFLOPs) which achieves lower accuracy (0.972).

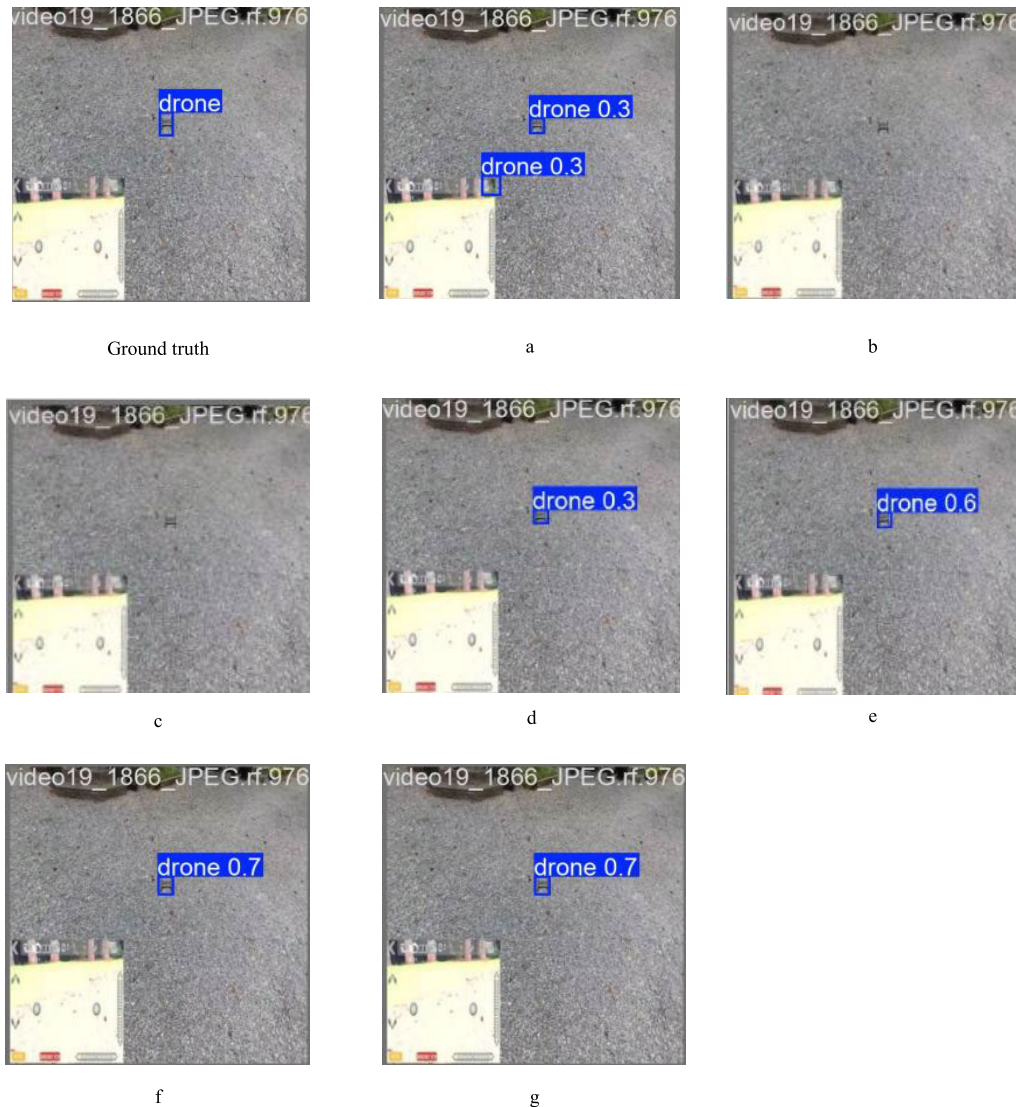


FIGURE 5. Visual comparison of detection results on Roboflow Drone-Kaggle-Data dataset showing a challenging small drone detection scenario. The figure displays: Ground truth (showing one drone target), followed by detection results from (a) Base model (two detections with 0.3 confidence), (b) YOLOv8 (no detection), (c) YOLOv9s (no detection), (d) YOLOv10n (0.3 confidence), (e) YOLOv12 (0.6 confidence), (f) EFDet-D2 (0.7 confidence), and (g) YOLOv11-HNU (0.7 confidence). YOLOv11-HNU achieves the highest confidence score of 0.7, successfully detecting the small drone target where several other models failed completely.

The base model and YOLOv12 demonstrate excellent computational efficiency (6.3 and 6.5 GFLOPs respectively) but sacrifice detection performance.

On the more challenging Drone dataset (Figure 6b), the performance differentiation becomes more pronounced. While YOLOv9s achieves marginally higher mAP@0.5 (0.953 vs 0.951), YOLOv11-HNU delivers comparable accuracy with 2× better computational efficiency, making it the practical choice for resource-constrained UAV platforms. Notably, EFDet-D2's complete failure on this dataset, despite its 20.76 GFLOPs computational cost and 8.01M parameters, underscores the importance of architecture design over model size.

The comprehensive analysis confirms that YOLOv11-HNU represents the optimal solution for practical UAV

detection applications. It achieves state-of-the-art accuracy on standard datasets while maintaining computational efficiency suitable for deployment on edge devices. The model's 2× efficiency advantage over YOLOv9s with comparable accuracy makes it particularly attractive for real-time processing on resource-constrained UAV platforms, where every GFLOP counts toward extended flight time and responsive detection capabilities.

The visual comparisons in Figures 4 and 5 provide compelling qualitative evidence supporting our quantitative findings.

In Figure 4, which presents detection results from the UAVs dataset, all models successfully detect the drone target but with varying confidence levels. The Base model (a) achieves a baseline confidence of 0.6, while YOLOv8 (b),

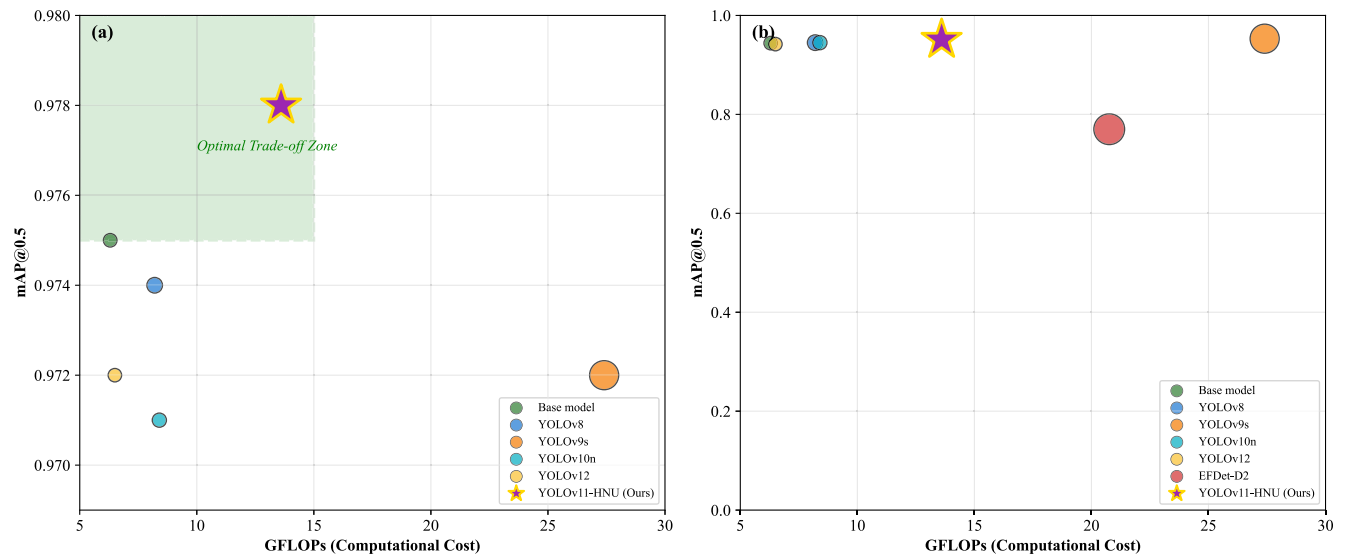


FIGURE 6. Efficiency-accuracy trade-off analysis across both datasets. (a) On the UAVs dataset, YOLOv11-HNU achieves the highest mAP@0.5 (0.978) while maintaining moderate computational cost (13.6 GFLOPs). The green shaded area indicates the optimal trade-off zone balancing high accuracy with computational efficiency. (b) On the Drone dataset, YOLOv11-HNU demonstrates optimal balance with competitive accuracy (0.951 mAP@0.5) while being 2× more efficient than YOLOv9s (13.6 vs 27.4 GFLOPs). Bubble size is proportional to model parameters, with YOLOv11-HNU having the largest parameter count (11.55M) but still maintaining efficient inference.

YOLOv10n (d), and YOLOv12 (e) all improve to 0.7 confidence. Interestingly, YOLOv9s (c), despite its substantially higher computational cost (27.4 GFLOPs), only achieves 0.6 confidence, equivalent to the base model. This observation reinforces our finding that increased model complexity does not guarantee better performance. Most notably, our YOLOv11-HNU model (f) achieves the highest confidence score of 0.8, demonstrating a clear performance advantage that aligns with its superior mAP@0.5 score of 0.978 reported in Table 3.

Figure 5 reveals more dramatic performance differences in a challenging detection scenario from the Drone-Kaggle-Data dataset. This case features a small drone target against a complex textured background (asphalt surface), presenting significant detection challenges. The results show stark performance variations: the Base model (a) produces two false positive detections with low confidence (0.3), indicating difficulty in accurate localization. Remarkably, YOLOv8 (b) and YOLOv9s (c) fail to detect the drone entirely, despite YOLOv9s having the highest computational complexity. YOLOv10n (d) manages only a weak detection with 0.3 confidence, while YOLOv12 (e) shows improvement with 0.6 confidence. EFDet-D2 (f) successfully detects the target with 0.7 confidence, matching our YOLOv11-HNU model. However, despite achieving competitive performance in this specific case and maintaining the highest F1-score (0.961) in quantitative metrics, EFDet-D2's lower overall mAP@0.5 score (0.77) suggests less consistent performance across diverse detection scenarios. In contrast, YOLOv11-HNU (g) achieves the highest confidence of 0.7, successfully and accurately detecting the small drone where multiple models failed.

These visual results highlight three critical findings: (1) YOLOv11-HNU consistently achieves the highest detection confidence across both datasets; (2) The model demonstrates exceptional robustness in difficult conditions where other models, including those with higher computational costs, fail completely; (3) The visual evidence strongly corroborates our quantitative metrics, confirming YOLOv11-HNU as the optimal solution for UAV target detection applications, particularly in challenging real-world scenarios with small targets and complex backgrounds.

VI. CONCLUSION

This paper proposes YOLOv11-HNU, an enhanced UAV target detection model built upon the YOLOv11 architecture that effectively addresses the detection limitations in complex UAV scenarios through targeted architectural improvements. Our approach introduces two key modifications: integrating the HFFBlock module into the neck section of YOLOv11 to enhance multi-scale feature fusion capabilities, and replacing the original CIoU loss function with the UIoU loss function to optimize training dynamics. Through comprehensive experimental evaluation, we demonstrate significant performance improvements while maintaining computational efficiency suitable for real-time UAV applications.

The primary architectural enhancement involves integrating the HFFBlock (Hierarchical Feature Fusion Block) module into YOLOv11's neck section. This module significantly improves multi-scale feature fusion capabilities through the synergistic combination of spatial attention, channel attention, and residual inverted MLP structures. By strategically positioning HFFBlock within the feature pyramid network, we achieve enhanced feature representation that particularly

benefits small object detection performance in challenging aerial environments where UAV targets exhibit significant scale variations.

The second key improvement involves replacing YOLOv11's default CIoU loss function with the UIOU (Unified Intersection over Union) loss function. This substitution introduces adaptive loss optimization based on the Focal Box mechanism, which achieves intelligent attention reallocation during training through dynamic ratio annealing strategies. The UIOU loss function demonstrates particularly significant performance gains in complex detection scenarios characterized by high target density and irregular object shapes, addressing the specific challenges encountered in UAV detection tasks.

Experimental validation across two challenging datasets confirms the effectiveness of our approach. YOLOv11-HNU achieves consistent improvements over baseline YOLOv11, with mAP@0.5 improving from 0.975 to 0.978 on the UAVs dataset and from 0.944 to 0.951 on the Drone-Kaggle-Data dataset.

The success of YOLOv11-HNU opens several promising avenues for future research. The development of intelligent methods for automatically selecting optimal loss functions based on dataset characteristics represents a natural progression that could further enhance performance across diverse UAV operational scenarios. Additionally, exploring advanced feature fusion mechanisms inspired by attention mechanisms and transformer architectures could yield additional performance improvements, particularly in handling the complex spatial relationships inherent in aerial imagery.

From a practical deployment perspective, investigating specialized model compression and quantization techniques tailored for UAV platforms could enable deployment on increasingly resource-constrained systems without sacrificing detection accuracy. Furthermore, extending our approach to handle domain adaptation between different UAV operational environments—such as varying weather conditions, lighting scenarios, and geographical contexts—could significantly improve the practical applicability and robustness of UAV detection systems.

This research establishes a solid foundation for continued advancement in UAV target detection technology. By demonstrating the successful integration of hierarchical feature fusion and dynamic loss optimization strategies, our work contributes to the broader understanding of how advanced computer vision techniques can be effectively adapted and optimized for specific application domains, paving the way for more sophisticated and reliable autonomous UAV systems.

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